Flaws In Plate Glass

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Abstract

We examine models for determining the theory of bubble nucleation and growth, and the movement of bubbles in molten glass. This work, when extended, will enable plate glass manufactures to improve the quality of plate glass.

1 Introduction

PGF Building Glass is a South African company that produces plate glass for the local building industry and also windscreens for automobiles. The company also exports to international markets.

Flaws occur as bubbles in the glass as seen in figure 1. Our aim is to understand the processes that occur in the bubble formation. Once we understand these processes, it will help us to improve the quality of the glass produced and reduce the loss.

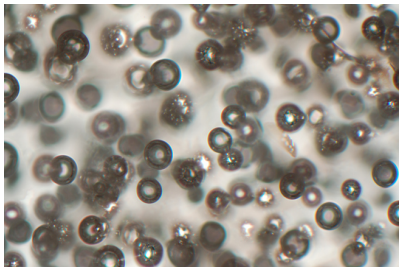


Figure 1: *The bubbles in a furnace during glass production*

Bubbles greater in size than 0.03 mm will affect the optical properties of the product. Such bubbles are formed in the furnace and need to be removed before leaving it. If glass plate is contain more than a prescribed number of such bubbles per unit area it will be discarded and recycled. Recycling is a very costly process.

Understanding how the bubbles are formed, move and grow in the furnace will help making informed decision on what to do in order to reduce or completely eliminate bubbles in the glass.

Molten glass is produced by radiating a mixture of sand (silica SiO_2) and limestone in a furnace. The relevant chemical equation is:

$$nSiO_2 + Na_2CO_3 \Leftrightarrow Na_2O - nSiO_2 + CO_2 \Uparrow \Delta H_{chem}. \tag{1}$$

This reaction produces a lot of carbon-dioxide which appears in bubbles. Also nitrogen (N_2), enters into the chamber from the air accompanying the batch. These two gases are relatively insoluble in molten glass, and therefore they form bubbles. Other gases are found in the furnace (like Na_2O) but these gasses are soluble in the glass and they do not produce bubbles.

Most of the formed bubbles rise to the top due to buoyancy and burst reducing bubbles content by factor of 10^7. The large ones rise faster but the very small ones rise slowly and are carried along with the flow and remain in the glass. The aim is to remove the few small remaining bubbles with radius about 0.03 mm which affect the quality of the glass. A method that is environmentally safe is desirable.

There are two methods that are currently being used to remove the bubbles:

- Adding fining agents
- Introducing larger bubbles at the bottom of the chamber.

Both methods produce more (large) bubbles that act by sweeping away the smaller ones.

In section 1 we describe aspects of furnace operation, and in section 2 we discuss the formation and the physics of bubble growth. Later in section 3 we describe the motion of the bubbles in the furnace, finally, we briefly describe procedures used to remove the few remaining small bubbles.

2 Furnace Operation

A batch typically consists of 70% silica, 14% soda ash and 10% limestone is feed to the furnace (40 m length, 10 m width and 4 m depth).

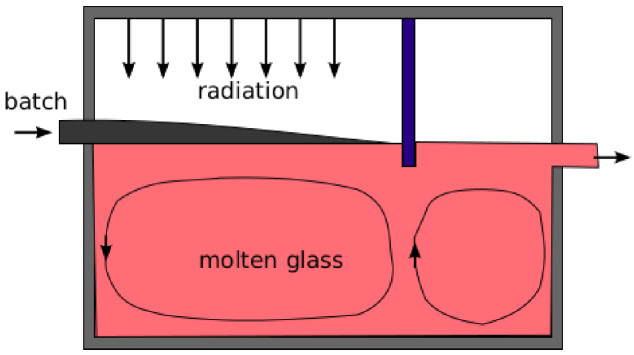


Figure 2: *the glass furnace. The batch is melted by the radiation from the ceiling.*

The batch is melted using radiation and this produces liquid glass, which is fed into the already molten glass see Figure 2. The temperature of the furnace is approximately 1500 °C and the particles takes about $10-15$ hours within the bath. Excess glass flows into the second chamber as shown in Figure 2 and this chamber is kept at a lower temperature (1400 °C). The molten glass then flows into a third chamber which consists of a flat thin tray (10 mm high) containing molten tin. The molten glass spread out over the tin to form a thin sheet of glass which solidifies producing plate glass. Any bubbles leaving the second chamber are frozen and remain in the plate glass. This process is called the Pilkington process.

3 The Physics of bubble formation

3.1 Nucleation

Nucleation is the localized formation of a distinct thermodynamic phase. In a liquid it is the formation of gaseous bubbles. There are two types of nucleation processes:

1. **Homogeneous nucleation** which occurs spontaneously and randomly in the liquid and requires supersaturation. It is hard to achieve under unsaturated conditions, even if the bubbles are formed they need to have a radius greater than the critical radius for them to survive, otherwise they shrink and disappear.

2. **Heterogeneous nucleation** is a continuous flow of bubbles occurs at nucleation site on surfaces contacting the liquid. Thus the gas travels along the surface and is transferred into the cavity creating a gas pocket, and when it is big enough it releases a bubble into the fluid. The radius of the cavity needs to be larger than the critical radius for the bubbles to be formed.

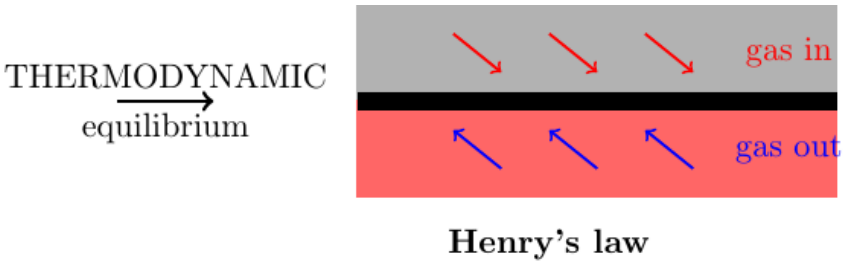
Except under supersaturation conditions only heterogeneous nucleation can occur (eg, bubbles in a glass of soda pops). In the molten glass case heterogeneous nucleation will occur unless fines are added.

3.2 Individual Bubble Behaviour

To understand how the bubble behave in the molten glass a single bubble model is considered. The bubble grows due to the transfer of gas across the surface of the bubble. Henry's law tells us about the transfer of flux of gas across the surface. We apply it in the bubble situation which show us that the concentration of gas in a liquid (c_l) is proportional to the partial pressure (P) of the gas above the liquid.

$$c_l = H(T)P \tag{2}$$

where H is the Henry's constant which is temperature dependant. This law only applies at thermodynamic equilibrium.



3.3 Laplace equation

This law applies for flat surfaces and need to be modified to take into account surface tension effects when dealing with bubbles. The pressure inside a bubble (p_b) is greater than that outside the bubble (p_l) due to surface tension (γ):

$$p_b = p_l + \frac{2\gamma}{R}, \tag{3}$$

a result referred to as Laplace equation (3), where R is the radius of the bubble. Note that surface tension effect are very large for small bubbles, see Figure (3). In fact bubbles with radius less than the critical radius cannot be formed in the body of the liquid.

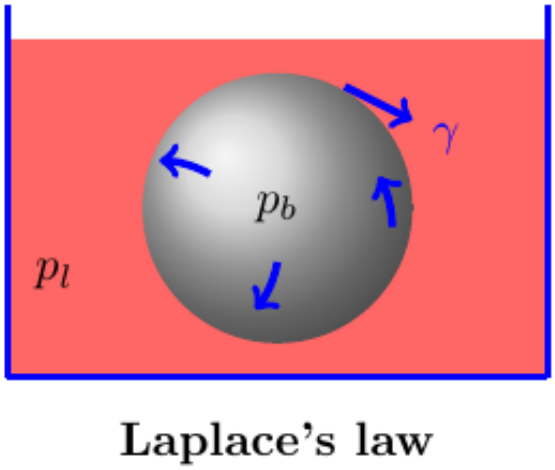


Figure 3: Surface tension effects

Small bubbles have a huge amount of pressure but a large bubble's pressure is almost equal to the pressure of the liquid.

3.4 Critical radius

The critical radius is the implication of the two equations (3) and (2), it is an equilibrium situation which can arise. The critical radius is given by:

$$R_c = \frac{2\gamma}{\frac{c_l}{H} - p_l} \tag{4}$$

This equilibrium is unstable. Where if the radius of the bubble is greater than the critical radius ($R > R_c$) the bubble will growth, else if the radius is less than the critical radius ($R < R_c$) it will shrink and disappear, as is showing in figure (4).

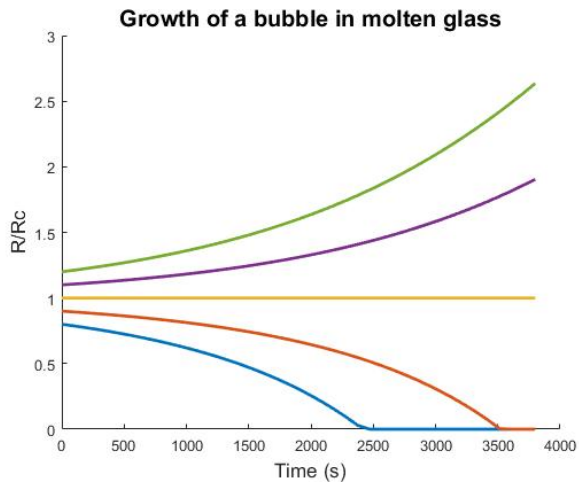
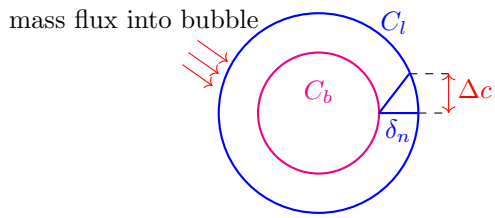


Figure 4: How bubble behave in different starting radius

The graph show us that when the radius of the bubble is bigger, the bubble will growth faster, and when it is smaller than the critical radius it will shrink and disappear faster.

3.5 The bubble growth

Given that the surface of a bubble is closed, the change in time of the mass enclosed inside the bubble is equal to the mass that traverses the surface.



This is the law of mass conservation:

$$\frac{d}{dt}\left(\frac{4}{3}\pi R^3 \rho_{b(t)}\right) = k4\pi R^2 (C_l - C_{b(t)}) \tag{5}$$

Using the chain rule, algebraic computations lead to the bubble growth equation:

$$\frac{dR}{dt} = \frac{\frac{k}{\rho_0}(C_l - C_b(t))}{1 + \frac{2\varepsilon}{3R}}$$

Applying the Laplace equation, we obtain

$$\frac{dR}{dt} = \frac{\frac{k}{\rho_0}\left(c_l - H\left(p_l + \frac{2\gamma}{R}\right)\right)}{1 + \frac{2\epsilon}{3R}}, \tag{6}$$

which matches to our understanding of the critical radius. Note that the right hand side of this equation vanishes when the radius equal to the critical radius ($R = R_c$).

4 Movement of the bubble

4.1 Without background flow

The movement of the bubble if we assume there is no convectional flow carrying the bubbles in the flow, buoyancy effects cause a bubble to move upwards The formation of bubbles displace the liquid around it, the buoyancy force which is applied by the fluid will try to lift the bubbles upward. This relation is represented in by equation:

$$F_{buoyancy} = \frac{4}{3}\pi R^3 \rho_s g \tag{7}$$

Assuming that the bubbles formed are spherical, and they are in a viscous fluid, the Stoke's drag force and gravity is try to pull the bubble down.

$$F_{drag} = 6\pi R \mu \dot{z} \tag{8}$$

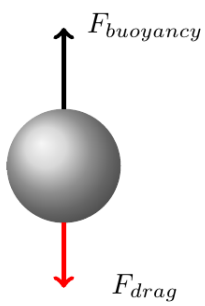


Figure 5: The acting force on a bubble

The acceleration of the bubble depend on the F_b and F_d by:

$$m\ddot{z} = F_{buoyancy} - F_{drag} \tag{9}$$

The density of the molten glass is very large compared to the mass of the bubble, from equation 9 we neglect its mass. At terminal velocity the two forces balances,

$$\begin{aligned} 6\pi R \mu \dot{z} &= \frac{4}{3}\pi R^3 \rho_s g \\ \implies \dot{z} &= \frac{2}{9}\frac{R^2(t)\rho_s g}{\mu} \end{aligned} \tag{10}$$

We now have two simultaneous equation (6) and (10) for determining the growth and movement of bubbles in the furnace. We run simulation of bubbles in the furnace using MATLAB to produce the result shown in Figure 6. Where bubbles are released in time interval while keeping the starting radius and the starting depth constant.

Recall that the molten furnace is 4 m depth and the bubbles are subjected to a haydrostate pressure which varies as they move ie, $p_l = \rho_M gh$, where h is the depth below the surface.

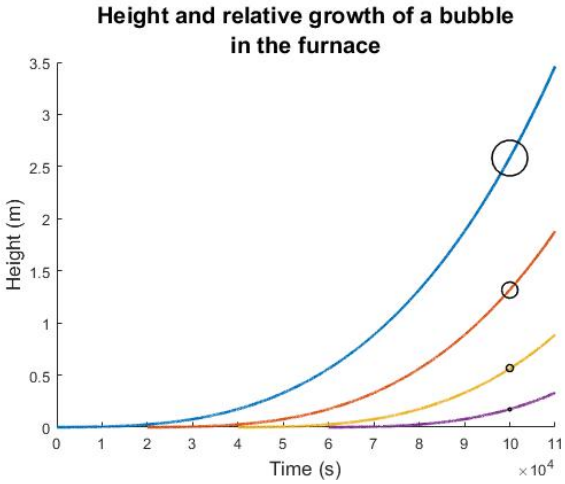
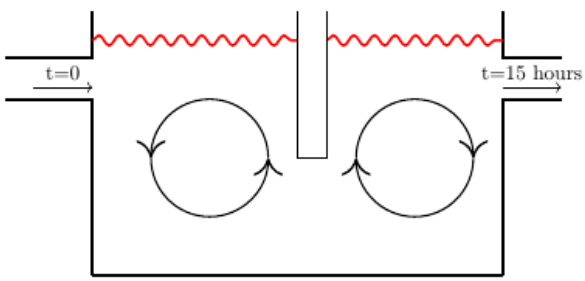


Figure 6: Numerical solution of bubbles growth

Note that the bubbles grow with time and accelerate.

4.2 In the presence of background flow

The furnace is very large and the temperature inside is very high and this causes convectional flow of the fluid in a circular motion with a speed of $3 \times 10^{-4}\, m/s$. The Bubbles are carried along with the flow but buoyancy effects cause the large bubbles to drift away from the flow and then reach the surface. But small bubbles don't drift and stuck in the flow and those are likely to remain in the chamber.



The relevant equation for the movement and growth of bubbles are the 2 ODEs (6 and 10) which need to be solved simultaneous.
There was insufficient time to complete calculations on these equations, however would expect an outcome similar to that in Figure(7),

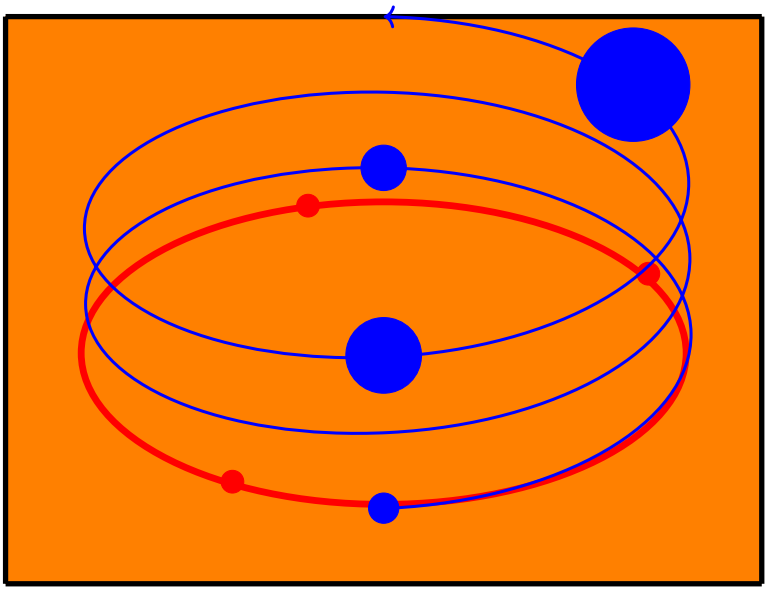


Figure 7: The movement and growth of bubbles

when bubbles with a radius bigger than the R_c (blue ones) growth and follows a spiral movement to reach the surface and shrink, when others (red ones) stuck in the flows.

5 Procedures to sweep up small bubbles

5.1 Fining agents:

Glass making companies use chemicals such as arsenic oxide to remove remaining small bubbles. These chemicals, called fining agents, are decomposed in the molten glass due to the high temperature. This results in diffusion of relevant gases into the growing bubbles and hence, in high ascension rates.

The arsenic oxide is one of the most efficient chemical fining agents, however it is highly toxic so that its use is undesirable To overcome the latter problem, we add alkali and alkaline earth oxides. Besides, less noxious compounds like sodium sulphate, sodium chloride and sodium nitrate have been used.

5.2 Introducing more bubbles:

Another way to sweep up small bubbles is to use a device that inserts bubbles into the molten glass. Those mechanically created bubbles have $10\,cm$ to $30\,cm$ diameter which is much above the critical radius and huge enough to absorb small bubbles. Mathematically, using these huge bubbles is more efficient than

introducing a swarm of smaller bubbles even though the last method increases the surface area.

6 Conclusion

The aim of this work was to a find better way to remove the few remaining small bubbles in glass. We investigated how a bubble grows and moves in the molten glass. As result, we found a critical range which gives us insight to the problem and leads us to a preliminary investigation to remove bubbles.

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